Comparing RNA models for something something

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Abstract

results indicate the entropy-ordered enumeration is a promising general-purpose technique to accelerate goal-directed top-down search for program synthesis.

I. Introduction

RNA molecules can be grouped into families based on shared characteristics such as structural patterns, sequence similarities, and functional roles. This allows researchers to study them in a more organized and meaningful way. To capture these similarities, a technique known as Multiple Read Alignment (MRA) is often employed, in which several RNA sequences from the same family are aligned together. From this alignment, a model can be constructed that encodes the common features of the family. This model can then be used as a reference to evaluate new RNA reads, providing a measure of how likely it is that a given sequence belongs to the same RNA family.

The chosen model for representing RNA families has often been a Stochastic Markov Model (SMM), which is capable of describing many RNA molecules in terms of both their raw symbolic strings and certain structural configurations that the molecules can take. This model has the same expressive power as a Stochastic Regular Grammar (SRG), meaning it can capture a wide range of sequence and structural patterns. However, it is limited in that it cannot represent more complex structures. These more intricate configurations are referred to as Secondary Structures (SS), and they require more powerful modeling techniques to be fully captured. To address this issue, a Covariance Model(CM) is often used to model these structures. The CMis a form of restricted Stochastic Context Free Grammar(SCFG) which captures a subset of the language of nested strings, the same set represented by Stochastic Visibly Pushdown Automaton(SVPA). **Bryan:** verify this, it is rather apparent, but still find a source.

As we are primarily concerned with secondary structural properties, we choose to use the latter technique when comparing different RNA families. We hypothesize that RNA families themselves can be examined for similarity, by comparing the similarity of the models used to represent them. We propose a modified graph edit distance for examining the similarity of different Covariance Model's which compares both structure and log-oddstransitions.

We validate our algorithm as faithfully finding the smallest edit distance between two Covariance Model's in II. To validate the efficacy of our approach we compare our results to the cliqueof RNA families. These cliques encode how similar families are to each-other based upon empirical evidence. **Bryan:** verify the name and purpose of clicks By comparing the distance between models, to their respective clique's we infer a cutoff and validate upon a hold out set. We present these results in **??**.

II. Formalism

A. Zhang-Shasha Algorithm

B. Hidden Markov Model Distance

III. Analasys

A. Time Complexity

B. Space Complexity

IV. Implementation

V. Evaluation

VI. Discussion

VII. Related Work

VIII. Conclusion

IX. Disclaimer

This report was written with assistance from an llm working in a strictly editorial capacity. All content and ideas are the original work of the authors.